

Pearson Edexcel International GCSE

Mathematics A

Modular Unit 2

June 2025 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

31 working units 106 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each page contains one main demand or one small inseparable chain. Source panels are followed by adaptive working space; writable diagrams count as working space.

Unit ownership rule: a question belongs to Unit 2 when its assessed, new or terminal statement is Unit 2, even when the working uses Unit 1 prerequisites. For example, the region-of-inequalities question remains Unit 2 although drawing and reading straight lines is supporting Unit 1 knowledge.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

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- 1 The table gives information about the distances 100 adults travel to work.

Distance (d km)	Frequency
$0 < d \leq 5$	26
$5 < d \leq 10$	40
$10 < d \leq 15$	16
$15 < d \leq 20$	10
$20 < d \leq 25$	8

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean distance.

..... km
(4)

WORKING SPACE

- 2 Anna makes cups.
Each cup costs 6 Swiss francs to make.

Anna puts the cups into boxes to sell.
Each box contains 4 cups.

Anna sells 80 boxes of cups for a total of 2160 Swiss francs.

- (a) Work out the percentage profit Anna makes.
Show your working clearly.

..... %
(4)

WORKING SPACE

3

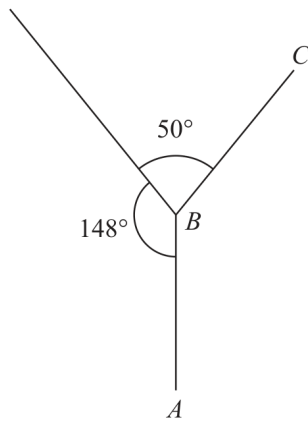


Diagram **NOT**
accurately drawn

AB and BC are two sides of a regular polygon with n sides.

Work out the value of n
Show your working clearly.

$n = \dots\dots\dots$

(Total for Question 3 is 4 marks)

WORKING SPACE

- 5 In a sale, normal prices are reduced by 28%
The sale price of a watch is 198 euros.
Work out the normal price of the watch.

..... euros

(Total for Question 5 is 3 marks)

WORKING SPACE

7 Here is a solid cylinder.

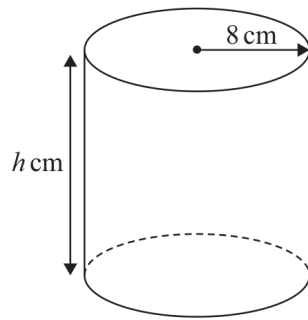


Diagram **NOT**
accurately drawn

The radius of the cylinder is 8 cm

The height of the cylinder is h cm

The volume of the cylinder is 3892 cm^3

Work out the value of h

Give your answer correct to one decimal place.

$h = \dots\dots\dots$

(Total for Question 7 is 3 marks)

WORKING SPACE

8 (a) Write 520 million in standard form.

.....
(1)

(b) Write 8.79×10^{-5} as an ordinary number.

.....
(1)

WORKING SPACE

- (c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

.....
(2)

WORKING SPACE

13 Charlotte bought an apartment for \$750 000

In the first year, the value of the apartment decreased by 4%

In the second year, the value of the apartment decreased by 6.5%

In the third year, the value of the apartment increased by $x\%$

At the end of the third year, the value of Charlotte's apartment was \$698 445

Work out the value of x

$x = \dots\dots\dots$

(Total for Question 13 is 3 marks)

WORKING SPACE

16 $f(x) = 9 - \sqrt{x}$ where $x \geq 0$

$g(x) = 4x^2$

(a) Find $f(9)$

(1)

WORKING SPACE

16 $f(x) = 9 - \sqrt{x}$ where $x \geq 0$
 $g(x) = 4x^2$

- (b) Solve $fg(x) < 0$
Show clear algebraic working.

.....
(3)

WORKING SPACE

18 y is inversely proportional to x^3

$$y = 4 \text{ when } x = 3$$

Work out the value of x when $y = 864$

$$x = \dots\dots\dots$$

(Total for Question 18 is 4 marks)

WORKING SPACE

19 The equation of a curve is $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are $(8, -12)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 3$

(.....,)
(1)

(ii) $y = f(2x)$

(.....,)
(1)

(Total for Question 19 is 2 marks)

WORKING SPACE

22 Solve the simultaneous equations

$$\begin{aligned}x^2 + 3y + y^2 &= 7 \\ y &= x + 2\end{aligned}$$

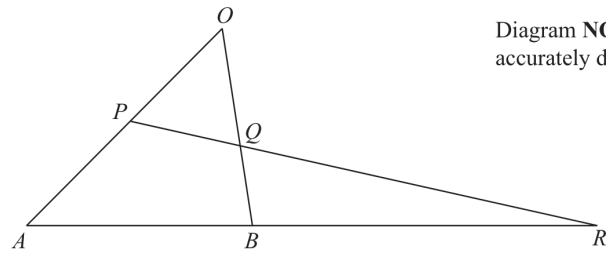
Show clear algebraic working.

.....

(Total for Question 22 is 5 marks)

WORKING SPACE

23

Diagram **NOT**
accurately drawn

OAB is a triangle.
 P is the midpoint of OA
 Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

.....
 (1)

WORKING SPACE

Continued
4MA1-1H Q23

UNIT 2

23

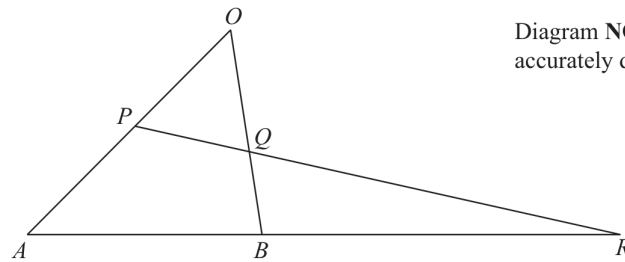


Diagram **NOT**
accurately drawn

OAB is a triangle.
 P is the midpoint of OA
 Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

$$n = \dots\dots\dots (4)$$

WORKING SPACE

24 An arithmetic series has 30 terms.

The first term is a

The common difference is d

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of a and the value of d

Show clear algebraic working.

$a =$

$d =$

(Total for Question 24 is 5 marks)

WORKING SPACE

- 1 Rio has six cards.
He writes a number on each card so that

the range of the numbers is 10
the median of the numbers is 7.5
the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14
--	--	---	--	----	----

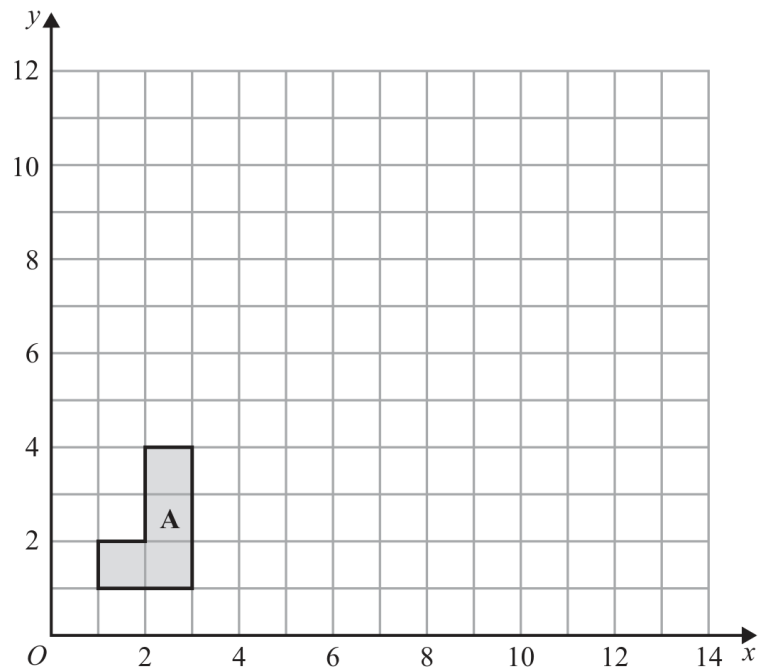
Three of the numbers are hidden.

Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

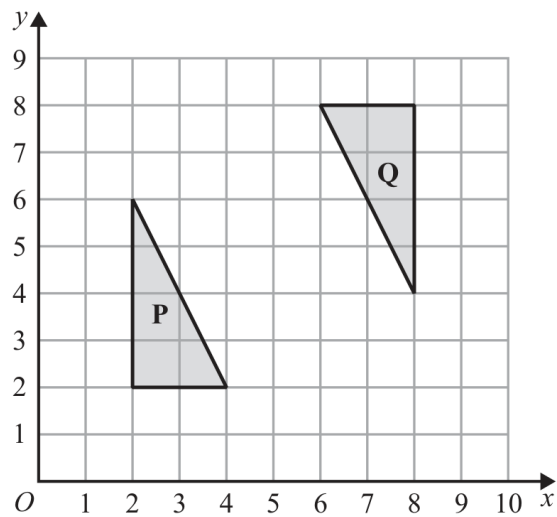
WORKING SPACE

2



(a) On the grid, enlarge shape A with scale factor 3 and centre $(0, 1)$

(2)



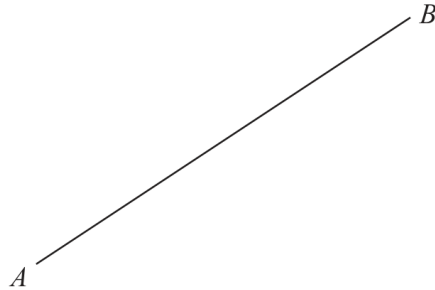
(b) Describe fully the single transformation that maps triangle **P** onto triangle **Q**

.....

.....

(3)

- 4 Using ruler and compasses only, construct the perpendicular bisector of the line AB .
Show all your construction lines.



(Total for Question 4 is 2 marks)

- 5 The diagram shows two similar quadrilaterals, $ABCD$ and $EFGH$

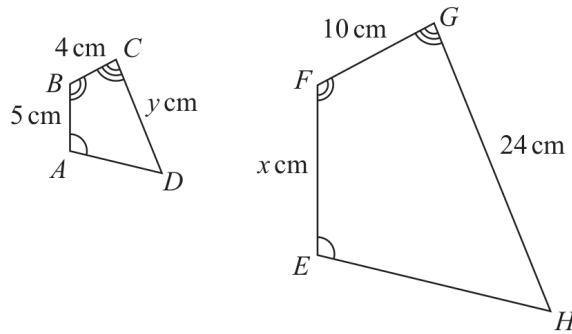


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm} \quad BC = 4 \text{ cm} \quad CD = y \text{ cm}$$

$$EF = x \text{ cm} \quad FG = 10 \text{ cm} \quad GH = 24 \text{ cm}$$

- (a) Work out the value of x

$$x = \dots\dots\dots (2)$$

WORKING SPACE

Continued
4MA1-2H Q5

UNIT 2

- 5 The diagram shows two similar quadrilaterals, $ABCD$ and $EFGH$

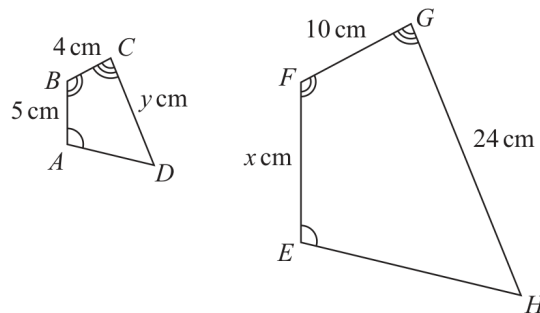


Diagram **NOT**
accurately drawn

$$\begin{array}{lll} AB = 5 \text{ cm} & BC = 4 \text{ cm} & CD = y \text{ cm} \\ EF = x \text{ cm} & FG = 10 \text{ cm} & GH = 24 \text{ cm} \end{array}$$

- (b) Work out the value of y

$$y = \dots\dots\dots (2)$$

WORKING SPACE

6 Pau, Sam and Tia share £240 in the ratios 3 : 4 : 5

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.
Give your answer in its simplest form.

..... : :

(Total for Question 6 is 4 marks)

WORKING SPACE

- 7 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.

Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

..... Swiss francs

(Total for Question 7 is 3 marks)

WORKING SPACE

8 Solve the simultaneous equations

$$\begin{aligned}3x + 5y &= 8 \\4x + y &= -3.5\end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 8 is 3 marks)

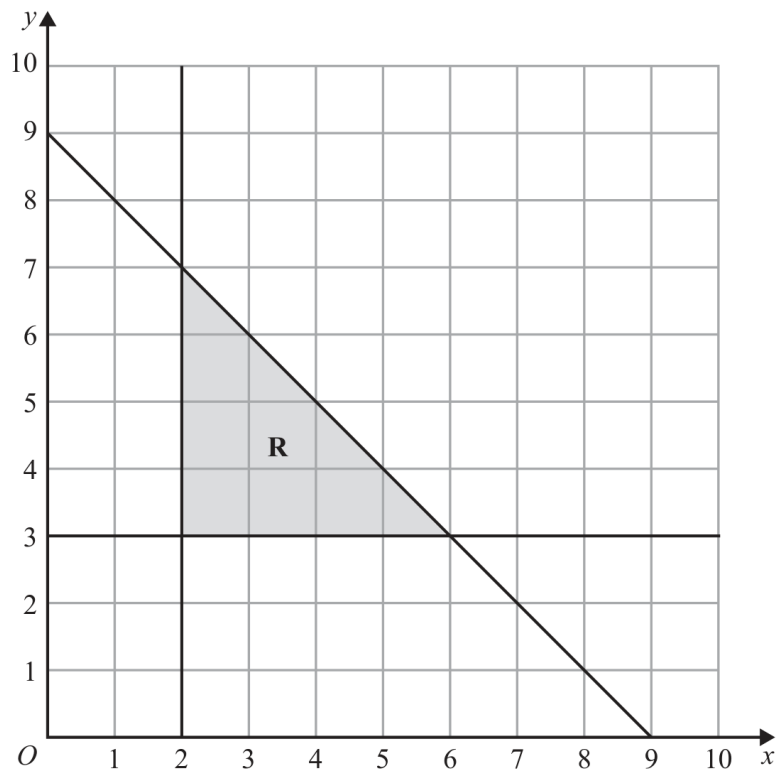
WORKING SPACE

9 (a) Solve the inequality $7 - 3t < 2t + 15$

(2)

WORKING SPACE

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down three inequalities that define the region **R**

.....

.....

.....

(3)

- 13 The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

Time (t minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	13
$20 < t \leq 30$	12
$30 < t \leq 40$	17
$40 < t \leq 50$	7
$50 < t \leq 60$	3

- (a) Complete the cumulative frequency table.

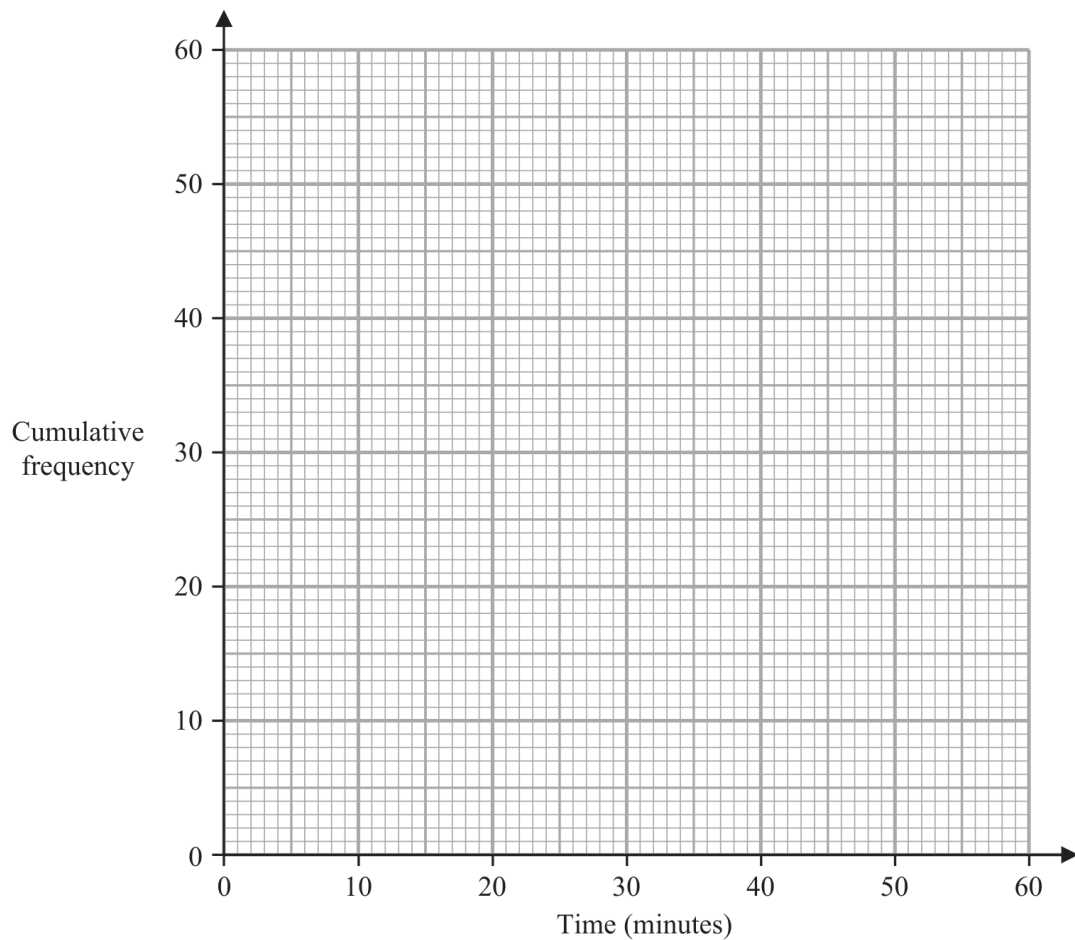
Time (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(1)

- (b) On the grid opposite, draw a cumulative frequency graph for your table.

Continued
4MA1-2H Q13

UNIT 2



(2)

- (c) Use your graph to find an estimate for the median time.

..... minutes
(1)

- (d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

.....
(2)

(Total for Question 13 is 6 marks)

16 Make t the subject of the formula $c = \frac{t^2 + 3}{7 - 8t^2}$

(Total for Question 16 is 4 marks)

WORKING SPACE

17 $y = 4x^3 + 5x^2 + 2x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

.....
(4)

(Total for Question 17 is 6 marks)

WORKING SPACE

- 19** Solve the inequality $4x^2 + 4x - 15 < 0$
Show clear algebraic working.

.....
(Total for Question 19 is 3 marks)

WORKING SPACE

- 21 The diagram shows a cuboid.

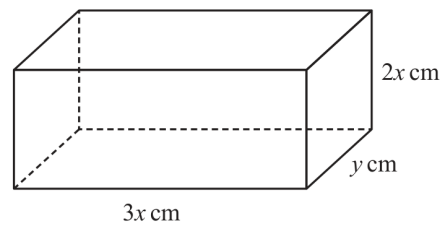


Diagram **NOT**
accurately drawn

The cuboid measures $3x \text{ cm}$ by $2x \text{ cm}$ by $y \text{ cm}$

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is $A \text{ cm}^2$

Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

WORKING SPACE

26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 26 is 4 marks)

WORKING SPACE